

# Lab 3: Central Tendency

## Sleep, Test Anxiety, and Memory During Exam Week

2026-08-11

### Student Information

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Estimated time:** 45–60 minutes

**Deliverables:** completed responses, SPSS output, a saved .sav file, and an APA-style results paragraph.

### Research scenario

A psychology research team is studying student well-being during exam week. They recruited 24 undergraduate students and recorded each participant's self-reported sleep from the prior night, test-anxiety score, performance on a short memory-recall task, and primary coping strategy.

The team's research question is:

**During exam week, what are the typical levels of sleep, test anxiety, and memory recall among undergraduate students, and which coping strategy is most common?**

This is a **descriptive** question. Your goal is not to test a hypothesis or compare groups. Instead, you will use the mean, median, and mode to describe the sample appropriately.

## Learning objectives

By the end of this lab, you should be able to:

1. Select the mean, median, or mode that best represents a variable.
2. Use SPSS to create frequency tables, descriptive statistics, and histograms.
3. Explain how an extreme score influences the mean and median.
4. Write a concise APA-style descriptive-results paragraph.

## Part A — Set up the psychology dataset

Download [the psychology lab dataset](#). In SPSS, select **File > Import Data > CSV Data** (or **File > Open > Data**, depending on your version). Confirm that variable names are read from the first row.

In **Variable View**, enter the value labels and measurement levels below.

| Variable        | Description / labels                                           | Measure |
|-----------------|----------------------------------------------------------------|---------|
| id              | Participant identification number                              | Nominal |
| coping_strategy | 1 = Mindfulness; 2 = Planning; 3 = Social support              | Nominal |
| sleep_hours     | Hours slept the night before the exam                          | Scale   |
| test_anxiety    | Test-anxiety score; possible range 0–40, higher = more anxiety | Scale   |
| memory_recall   | Number of words recalled correctly; possible range 0–20        | Scale   |

### Checkpoint A

Save the file as `lab_3_yourlastname.sav`.

**Question:** Why is `coping_strategy` nominal, even though SPSS displays it using the numbers 1, 2, and 3?

**Response:**

## Part B — The mode: Most Common Coping Strategy

Use **Analyze > Descriptive Statistics > Frequencies**. Move `coping_strategy` to the Variable(s) box. Click **Statistics**, select **Mode**, click continue, then OK. Keep the frequency table in your output.

1. Which coping strategy is the mode? Report its frequency and percentage.

**Response:**

2. Why would it be inappropriate to calculate the mean of `coping_strategy`?

**Response:**

3. In plain language, write one sentence that answers the coping-strategy portion of the research question.

**Response:**

## Part C — Typical sleep and memory performance

Use **Analyze > Descriptive Statistics > Explore**. Put `sleep_hours` and `memory_recall` in the **Dependent List**. Click on **Statistics**. Ensure *Descriptives* and Percentiles are selected. Under percentiles, select *Quartiles* then click *Continue*. Next, click *Plots*, then select *Histogram*. Click *Continue* the *OK*. An output should popup.

1. Record the mean and median for each variable.

| Variable                   | Mean | Median |
|----------------------------|------|--------|
| <code>sleep_hours</code>   |      |        |
| <code>memory_recall</code> |      |        |

2. For each variable, describe the histogram as roughly symmetric, positively skewed, or negatively skewed.

**Sleep hours:**

**Memory recall:**

3. For each variable, decide whether the mean or median is the better single summary of a typical score. Briefly justify your choices.

**Sleep hours:**

**Memory recall:**

## Part D — Test Anxiety and an Extreme Score

Use **Analyze > Descriptive Statistics > Explore** to obtain the mean, median, and histogram for `test_anxiety`. Refer to the steps described in part C above if you need to.

1. Record the mean, median, minimum, and maximum.

| Mean | Median | Minimum | Maximum |
|------|--------|---------|---------|
|------|--------|---------|---------|

2. Does the distribution appear skewed? Identify the value that appears to be extreme, and state whether it is low or high relative to the other scores.

**Response:**

3. Which measure of central tendency—mean or median—best represents a typical test-anxiety score in this sample? Explain your answer using the effect of the extreme score.

**Response:**

### Examine the Effect of the Extreme Score

Use **Data > Select Cases**. Choose **If condition is satisfied**, enter `id < 24`, then click **continue**. Select **Filter out unselected cases**. You will notice that some cases are crossed out. Re-run the Explore analysis for `test_anxiety`.

4. Complete the table. Round to two decimal places.

| Cases included  | Mean | Median |
|-----------------|------|--------|
| All 24 cases    |      |        |
| Cases 1–23 only |      |        |
| Change          |      |        |

5. Which statistic changed more after the extreme score was excluded? What does this demonstrate about the mean and median?

**Response:**

## Part E — A simple linear transformation

Before doing this part, we need to Restore **All cases**. To do this, return to **Data > Select Cases > All cases > OK**. This will clear the conditional filtering applied earlier.

Now, imagine the research team revises the anxiety scale by adding 5 points to every participant's score. Use **Transform > Compute Variable**. Under *Target Variable*, write `anxiety_plus_5`. Under Numeric Expression write `test_anxiety + 5`. then click **OK**. A new variable named `anxiety_plus_5` will be created.

Run Explore for `test_anxiety` and `anxiety_plus_5`.

1. Complete the table below:

| Variable                    | Mean | Median |
|-----------------------------|------|--------|
| <code>test_anxiety</code>   |      |        |
| <code>anxiety_plus_5</code> |      |        |

2. How did the mean and median anxiety change after adding 5 to each individual score? State the rule for what happens to the mean and median when the same constant is added to every score.

**Response:**

## Part F — Interpret and report the results in APA style

APA-style results writing should be accurate, concise, and tied to the research question. Use the statistics you produced in SPSS—not estimates from the graph.

### APA reporting reminders

- Report the sample size as  $N$  when referring to the whole sample.
- Report the mean as  $M$  and the standard deviation as  $SD$ .
- Report the median as  $Mdn$  when it is the better summary for a skewed distribution.
- Report a categorical mode in words, followed by its frequency ( $n$ ) and percentage when useful.
- Round consistently, typically to two decimal places unless your instructor directs otherwise.
- Describe the distribution when that explains why you selected the median instead of the mean.

### Build your results paragraph

Use this template to write a paragraph that answers the research question:

The sample ( $N =$  ) *reported an average of  $M =$  hours of sleep ( $SD =$  ;  $Mdn =$  ). Participants recalled an average of  $M =$  words ( $SD =$  ) *on the memory task. The modal coping strategy was [strategy] ( $n =$  , %). Test-anxiety scores were [roughly symmetric / positively skewed / negatively skewed]; therefore, the [mean / median] was the most appropriate indicator of a typical score ( $M$  or  $Mdn =$  ).* Overall, these results suggest that [one sentence summarizing the sample descriptively without making a causal claim].*

### Your APA-style results paragraph

Write a polished 4–6 sentence paragraph below. Use the correct statistical notation and do not state that one variable *caused* another; this study only describes the sample.

**Results:** \_\_\_\_\_

### Submission checklist

- ☐ A saved SPSS data file with correct value labels and measurement levels.
- ☐ SPSS output containing the frequency table, descriptive statistics, and histograms.
- ☐ Completed responses for Parts A–F. You may print a PDF from the website and hand-write your responses or use a digital annotator tool.

- One APA-style results paragraph that directly answers the research question.

 Optional challenge

Use **Graphs > Chart Builder** to make a bar chart of `coping_strategy`. Add an informative title and axis labels. In one sentence, explain how the tallest bar communicates the same information as the mode.